Predicting Player Market Value in FC 26

Phase 1 Report — Data Understanding & Preprocessing

PROBLEM STATEMENT

The market value of a football player is shaped by a wide range of attributes like skill ratings, physical profile, club and league context, and reputation, making it difficult to estimate through simple heuristics. This project aims to build a regression model that predicts a player's market value (value_eur) using structured attribute data from the FC 26 player database, providing a data-driven benchmark for scouting, transfer analysis, and squad valuation.

DATASET SOURCE & JUSTIFICATION

Source: FC 26 (FIFA 26) Player Data — Kaggle

This dataset was chosen because it provides an extensive, up-to-date snapshot of professional players' attributes (ratings, physical traits, club/league metadata, and market/release-clause values) sourced directly from the game's database, which mirrors real-world scouting data. Its size and attribute richness make it well suited for supervised regression on market value.

DATASET OVERVIEW

18 relevant columns were selected from the raw dataset, covering identity (long_name), attributes (age, height_cm, overall, potential, skill_moves, weak_foot, international_reputation, body_type, preferred_foot, work_rate), positional/club context (player_positions, club_name, club_position, league_name, league_level), and financials (value_eur which is the target, and release_clause_eur).

PREPROCESSING STEPS PERFORMED

- Selected 18 relevant columns from the raw FC26.csv export and dropped duplicate rows.
- Checked for missing values; dropped the work_rate column entirely as it was empty for all records.
- Fixed missing release_clause_eur values as 1.25 × value_eur, based on the typical release-clause premium.
- Simplified player_positions to a single primary position by splitting on commas and keeping the first entry.
- Dropped the long_name column (identifier, not predictive) before encoding.
- Label-encoded all non-numeric columns (positions, club, league, foot, body type, etc.).
- Standard-scaled the age column using StandardScaler.
- Built a correlation heatmap to inspect feature relationships and guide feature selection.
- Dropped low-relevance columns (age, height_cm, preferred_foot, league_name, player_positions) after correlation review.
- Removed outliers in league_level using the IQR method (boxplots before/after).
- Applied a log1p transform to value_eur (log_value) to correct its right-skewed, gamma-like distribution, since resampling methods (SMOTE/Tomek) do not apply to a continuous target.

LIBRARIES & TOOLS USED

Python, pandas, NumPy, seaborn, matplotlib, scikit-learn (LabelEncoder, StandardScaler), imbalanced-learn (SMOTE, TomekLinks, SMOTETomek — evaluated but not applicable to a continuous target), and Jupyter Notebook.

CHALLENGES ENCOUNTERED

- Missing release_clause_eur values had no direct substitute, resolved via a value-based estimation rule.
- Players could play in multiple positions, so only one was selected.
- Cannot remove outliers in most columns because they are the feature of our data.
- The target variable's continuous nature meant standard class-imbalance techniques (SMOTE/undersampling) could not be applied, since they assume discrete class labels.
- value_eur was heavily right-skewed (a small number of elite players with very high values), which could bias the model toward outliers; a log transform was applied to stabilize the feature-target relationship.

Project repository: github.com/S4lmanAli/PlayerMarketValuePredictor